

National University of Computer and Emerging Sciences



Lab Manual 09 **Computer Organization and Assembly Language Lab**

Department of Computer Science

FAST-NU, Lahore, Pakistan

P 1 Write a program that prints and prints following messages: one message per keypress.

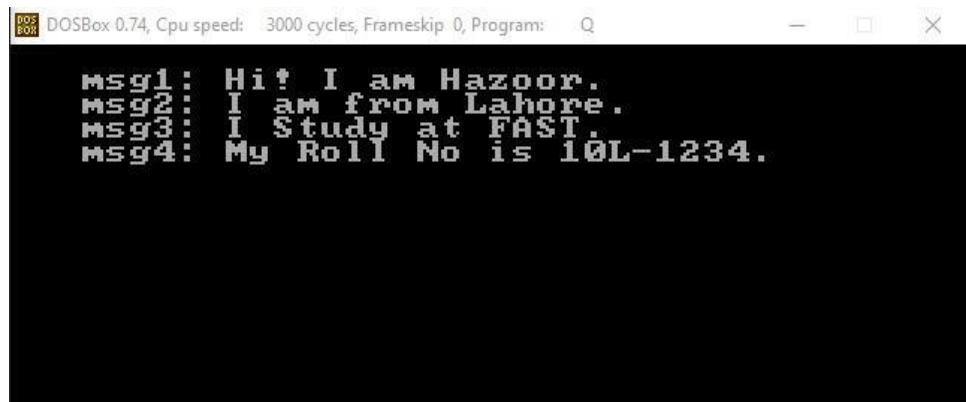
'msg1: Hi! I am YourName.'

'msg2: I am from YourCity.'

'msg3: I Study at FAST.'

'msg4: My Roll No is YourRoll#.'

Expected output after 4 key press

A screenshot of a DOSBox window. The title bar reads "DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program: Q". The window contains a black terminal area with white text. The text displayed is:
msg1: Hi! I am Hazoor.
msg2: I am from Lahore.
msg3: I Study at FAST.
msg4: My Roll No is 10L-1234.
The text is aligned to the left and each message is on a new line.

P 2 The following code segment prints a diagonal line. Modify this code to print the diamond as shown below.

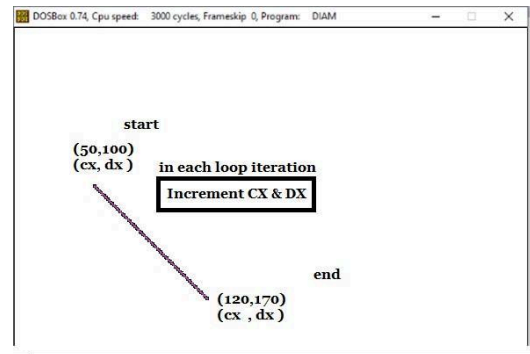
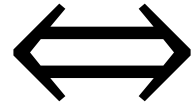
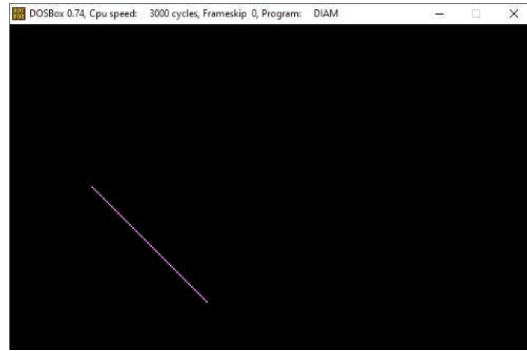
```

[org 0x0100]
jmp code
w equ 70      ; width offset x equ 50      ;
starting x coordinate of line y equ 100    ;
starting y coordinate of line
c equ 60      ; color
code: mov ah, 0
mov al, 13h    int
10h
; draw diagonal 11:
mov cx, x
    mov dx, y
    mov al, c u1:
    inc dx
    mov ah, 0ch ; put pixel
    int 10h
    inc cx
    cmp cx, x+w
    jbe u1
;wait for keypress
mov ah,00    int
16h
mov ax, 0x4c00
int 21h

```

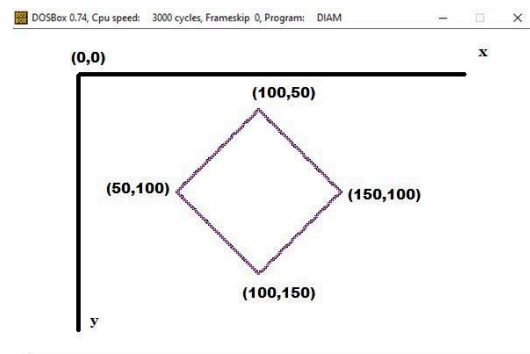
Given:

Start with given $(x, y) = (50, 100)$ and in every loop iteration increment both Cx and Dx to reach $(x + w, y + w) = (120, 170)$.



Required:

Similarly, you need to draw 4 lines as above satisfying the coordinates and increment or decrement as per requirement of the line as shown below.



P 3 Modify the code segment provided in **Problem 2** or write your own code draw the following alphabets on every key press.

N E X T